

LESSON 112 (2.0) GROUND

Lesson Objective

During this lesson, the applicant will become familiar with the maneuvers to be performed in this stage. The learner will gain an understanding of the concepts and procedures of takeoffs, landings, traffic patterns, maneuvering during slow flight, power-off stalls, power-on stalls, emergency operations, and ground reference maneuvers. Additionally, the learner will be introduced to towered and non-towered airport operations, and radio communications.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on basic aspects of a traffic pattern
 - Basic questions on turns around a point, S-turns and rectangular course
- Student questions.

References

- Airplane Flying Handbook ([FAA-H-8083-3C](#))
- Florida Tech Aviation Maneuvers Manual

Instructor Guidance

- Ensure your student has had the chance to review the assigned sections of the Airplane Flying Handbook prior to this lesson. If they have not, address that expectation clearly — the volume of content in this lesson makes preparation essential.
- Utilize the telling and doing technique throughout. Introduce each maneuver conceptually, then walk the student through it procedurally, and where possible use chair flying or physical demonstration to reinforce the sequence before they fly it.
- Reiterate the importance of clearing turns before every maneuver and the positive exchange of flight controls. These habits must be established now and should be referenced every time a new maneuver is introduced from this lesson forward.
- Reinforce coordinated flight throughout all maneuvers by referencing both the turn coordinator and outside visual cues. Students who develop sloppy coordination habits early will carry them all the way to the checkride.
- Encourage smooth, coordinated control inputs at all times — avoid abrupt or aggressive corrections. Frame this as both a safety and a skill issue: an examiner will notice uncoordinated flight immediately.
- When covering the cross-controlled stall, be clear that this is an instructor-demonstration only maneuver. Emphasize the real-world scenario it represents — overshooting final and

attempting to correct with opposite rudder — so the student understands exactly why this is one of the most dangerous situations in the traffic pattern.

- When covering emergency operations, stress that the ABCs (Airspeed, Best landing area, Checklists) are memory items. The student should be able to recite them without hesitation. Use this lesson to begin building that expectation before the maneuver is practiced in the aircraft.
- Continuously evaluate traffic, terrain, and airspace awareness throughout all maneuver discussions — frame situational awareness as a mindset, not just a checklist item.
- Demonstrate common student errors for each maneuver and explain their consequences. Students who understand what can go wrong are far more likely to self-correct in the aircraft.

Lesson content

Normal and Crosswind Takeoffs

- In accordance with the most current version of the approved Maneuvers Manual.

Normal and Crosswind Landings

- In accordance with the most current version of the approved Maneuvers Manual.

Traffic Patterns

- In accordance with the most current version of the approved Maneuvers Manual.
- Left vs right pattern. Use X59 & X26 as examples.
- Review traffic pattern procedures in Melbourne.
 - Crosswind at vs before the VOR
 - Downwind over Sarno Rd
 - Base leg over Wickham
- Review radio calls in the pattern at a non-towered airport vs towered airport.

Maneuvering During Slow Flight

- In accordance with the most current version of the approved Maneuvers Manual.
- The objective of maneuvering in slow flight is to develop the pilot's ability to fly at low speeds and high AOAs. Through practice, the pilot becomes familiar with the feel, sound, and visual cues of flight in this regime, where there is a degraded response to control inputs and where it is more difficult to maintain a selected altitude

Stall Aerodynamics

- A stall is caused by exceeding the critical angle of attack, not by airspeed alone.
- The wing exceeds the critical AOA → airflow separation → loss of lift.
- Stalls can occur at any airspeed, attitude, or power setting.
- Indications of approaching stall
 - Decreasing control effectiveness ("mushy" controls)
 - Need for increasing back pressure to maintain altitude

- Stall warning horn / annunciation
- Buffet (airframe vibration)
- Nose-high attitude
- Decreasing airspeed trend
- Coordinated vs. uncoordinated flight indications
- Full stall characteristics
 - Abrupt loss of lift
 - Nose pitch down (natural aerodynamic tendency)
 - Possible wing drop (due to yaw/imbalance)
 - Reduced aileron effectiveness

Power-Off Stalls

- In accordance with the most current version of the approved Maneuvers Manual.
- The practice of power-off stalls is usually performed with normal landing approach conditions to simulate an accidental stall occurring during approach to landing. However, power-off stalls should be practiced at all flap settings to ensure familiarity with handling arising from mechanical failures, icing, or other abnormal situations. Airspeed in excess of the normal approach speed should not be carried into a stall entry since it could result in an abnormally nose-high attitude.

Power-On Stalls

- A power-on stall simulates a stall during takeoff or departure when the aircraft is at high pitch with high power applied. As airspeed decreases and the critical angle of attack is exceeded, stall indications such as buffet, warning horn, and possible wing drop may occur.
- The maneuver emphasizes recognition of stall cues, maintaining coordination, and prompt recovery by reducing the angle of attack, leveling the wings, and establishing a positive climb.

The Base to Final Turn and Cross-Controlled Stalls (Instructor Demonstration)

- This maneuver is instructor-demonstration only and is conducted to reinforce stall/spin awareness, coordinated flight principles, and sound aeronautical decision-making in the traffic pattern.
- A cross-controlled stall occurs when the critical angle of attack is exceeded while the aircraft is uncoordinated, with aileron applied in one direction and opposite rudder applied in the other. This condition results in a skid or slip, indicated by the ball in the turn coordinator deflecting to the outside of the turn.
- This type of stall most commonly develops during the base-to-final turn when a pilot overshoots the runway centerline and attempts to correct by increasing bank angle and applying excessive rudder toward the runway. Combined with increased back pressure to maintain altitude, this uncoordinated condition increases the risk of an accelerated stall and potential spin entry at low altitude.

Emergency Operations

- Simulated Engine Failure
 - In accordance with the current and approved Florida Tech Aviation aircraft checklist.
 - Practiced on dual instructional flights only. The aircraft will not descend below 500 feet AGL.
 - Emphasize the ABCs:
 - **A**irspeed
 - **B**est landing area (select a suitable landing site)
 - **C**hecklists
 - Emergency checklists should be memorized. Review any missed items only if altitude and time permit.
 - Discuss suitable landing areas (e.g., farm fields, open roads, street patterns, airports) and areas to avoid (e.g., flooded fields, power lines, congested areas, and landing perpendicular to field rows).
- Review emergency checklist items from appropriate checklist

Turns Around a Point

- In accordance with the most current version of the approved Maneuvers Manual.
- Turns around a point are a continuation of previous ground reference maneuvers such as rectangular courses and S-turns. This maneuver consists of making a 360° turn around a selected reference point on the ground while maintaining a constant radius.
- The key principle is that bank angle must vary with groundspeed. When groundspeed is higher (downwind), a steeper bank is required. When groundspeed is lower (upwind), a shallower bank is required.
- To properly understand wind correction and bank adjustments, the pilot should complete at least one full 360° turn, though two turns are recommended for better wind assessment. Throughout the maneuver, the pilot must continuously adjust bank angle and wind correction to maintain equal distance from the reference point, using coordinated aileron and rudder inputs at all times.

S-Turns

- In accordance with the most current version of the approved Maneuvers Manual.
- An S-turn is a ground reference maneuver in which the airplane makes two equal 180° turns on opposite sides of a straight-line ground reference, such as a road. The goal is to maintain equal radius turns while correcting for wind drift.
- The maneuver is normally entered on the downwind side, where groundspeed is highest. After crossing the reference line at a 90° angle, the pilot immediately begins a 180° constant-radius turn. As the airplane moves from downwind to crosswind to upwind, groundspeed decreases, so the bank angle must gradually decrease to maintain a constant distance from the reference line.
- Upon crossing the reference line again, the pilot immediately begins a second 180° turn in the opposite direction. Since this turn begins upwind (where groundspeed is slowest), the initial bank angle should be shallow. As the airplane progresses from upwind to crosswind to downwind, groundspeed increases, so the bank angle must gradually increase to maintain a constant radius.
- The objective is smooth coordination and consistent ground track despite wind conditions.

- Throughout the maneuver, the pilot must:
 - Adjust bank angle to compensate for changing groundspeed
 - Apply proper wind correction angle
 - Maintain coordinated flight using aileron and rudder
 - Roll out on the reference line at the correct heading, altitude, and airspeed

Rectangular Course

- In accordance with the most current version of the approved Maneuvers Manual.
- The rectangular course is a ground reference maneuver in which the airplane maintains equal distance from all sides of a selected rectangular area. The purpose of this maneuver is to simulate the traffic pattern flown at an airport.
- During the maneuver, the pilot must maintain constant altitude, airspeed, and spacing from the ground references while correcting for wind drift.
- To perform the rectangular course, the pilot should select a safe area such as a square or rectangular field with clearly defined boundaries. The airplane should be flown parallel to each side of the field at a distance of approximately $\frac{1}{2}$ to $\frac{3}{4}$ mile, remaining outside the boundaries so the references are easily visible.
- The pilot should avoid flying too close to the boundaries, as this would require excessively steep bank angles, increasing load factor and stall speed—particularly during the downwind-to-crosswind turn.

Towered Airport Operations

- Obtaining taxi, takeoff, and landing clearances
- Compliance with all ATC instructions and readback requirements
- Traffic pattern entry and pattern compliance as instructed by the tower
- Runway incursion avoidance
- Airport signage and markings
 - Identify runway and taxiway markings
 - Interpret mandatory instruction, location, direction, and information signs
 - Recognize runway holding position markings
 - Understand taxiway centerline and edge markings
 - Identify displaced thresholds, aiming points, touchdown zone markings, and runway numbering
 - Recognize hotspots and surface painted holding position signs
- Situational awareness of other traffic.

Non-Towered Airport Operations

- Non-towered airport operations require pilots to self-announce intentions and maintain heightened situational awareness in the absence of ATC services.
- Students should understand and demonstrate:
 - Proper use of Common Traffic Advisory Frequency (CTAF)
 - Standard traffic pattern entry procedures (45° entry to downwind when appropriate)
 - Correct traffic pattern altitudes and left/right pattern compliance

- Clear and concise radio position reports (10 miles out, entering downwind, base, final, clear of runway, etc.)
- Visual scanning for traffic and collision avoidance
- Wind determination using windsocks, AWOS/ASOS, or traffic flow
- Emphasis should be placed on standard phraseology, predictable traffic pattern operations, and maintaining visual separation from other aircraft at all times.

Radio Communication Procedures and ATC Phraseology

- Students should understand and demonstrate:
 - Proper radio setup and frequency management
 - Initial call-up format (who you are calling, who you are, where you are, what you want)
 - Accurate readback of clearances and instructions (runway assignments, hold short instructions, altitudes, headings, squawk codes)
 - Use of standard terminology (e.g., "affirmative," "negative," "standby," "unable")
 - Avoidance of nonstandard or casual language
 - Maintaining situational awareness while communicating
- Emphasis should be placed on:
 - Thinking before keying the microphone
 - Speaking clearly at an appropriate pace
 - Listening fully before responding
 - Asking for clarification if instructions are unclear

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 107
- Assign homework:

Completion Standards

This lesson is complete when the applicant can:

1. Understand the concepts behind the maneuvers introduced.
2. Understand how to perform the maneuvers introduced.
3. Understand towered and non-towered airport operations.
4. Gain a basic understanding of radio communication procedures and ATC phraseology.